



Oluwayemi Oyinola Michael

Software Engineer · Backend Developer

Python & data engineering · distributed systems · application security

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— PROFESSIONAL SUMMARY

Backend-focused software engineer in Lagos, Nigeria, building production systems in Go, Python and TypeScript since 2023. I design and own the parts of a product users never see — APIs, data models, authentication, payments, queues, security controls and the Linux infrastructure underneath — from the first schema to the deployed service.

As founder and lead engineer of ZudoMart I designed a 79-module Go modular monolith across five bounded domains, with escrow, verification and risk scoring modelled as explicit state machines. I also author Zudojs, a 39-package open-source TypeScript framework, and have used it to ship multi-service systems such as BetNG — twelve TypeScript and Python services behind one 187-route gateway — alongside platforms for savings groups, schools, retailers, logistics and a local church.

Security runs through the work rather than beside it: SentinelX, a network intrusion detection and prevention platform with explained risk scores, and CommitGuard, a commit-provenance policy engine published to PyPI and the GitHub Marketplace. I write the tests, benchmarks and documentation that let another engineer trust a system without having to take my word for it.

Currently completing a BSc in Computer Science at the University of the People while deepening data engineering, distributed systems and application security. Open to backend, platform and software engineering roles.

— TECHNICAL SKILLS

Languages	Go · Python · TypeScript · JavaScript · SQL · Bash
Backend	Gin · Fastify · Express · FastAPI · Node.js · REST APIs · gRPC
Databases	PostgreSQL · MariaDB · MySQL · Redis · SQLite / libSQL
Data tooling	Prisma · ent + Atlas · sqlc · pgx · Sequelize · BullMQ · Kafka
Architecture	Modular monolith · Microservices · Event-driven · CQRS · DI containers · Hexagonal
Data	Data pipelines · Event processing · Ranking · Recommendations · Analytics
Security	Argon2 · TOTP / 2FA · JWT & sessions · RBAC · HMAC webhooks · Rate limiting · OWASP practice
Infrastructure	Linux · Kali · Docker · Git · SSH · Nginx · CI/CD · Terraform
Practice	API design · Swagger / OpenAPI · Postman · Testing · Documentation · CLI tooling
Frontend	React · Next.js · Tailwind CSS · Three.js · Vanilla HTML/CSS/JS

— SOFT SKILLS

Ownership & leadership	Founded ZudoMart and lead its engineering end to end; lead engineer on Telente CBT and LearnBridge.
Written communication	Design specifications, READMEs, API documentation and architecture write-ups that another engineer can build from.
Team collaboration	Ship on shared codebases with the Newdich team and an Orange internship team, through typed contracts and code review.
Problem solving & systems thinking	Break an ambiguous product idea into services, data models, workflows and failure cases before writing code.

Client & stakeholder communication	Delivered software for schools, a church, a food business and savings groups, translating their needs into scope.
Self-directed learning	Study for a CS degree alongside production work; taught myself Go, Python data tooling and security practice.
Remote work & time management	Work asynchronously across time zones from WAT (UTC+1), with milestone-driven delivery on multiple concurrent projects.

— PROFESSIONAL EXPERIENCE

Founder & Lead Engineer

2023 — present

ZudoMart

Africa-first social commerce platform combining marketplace, services, micro-gigs, live commerce and a multi-layer trust engine. Joined FasterCapital's EquityPilot programme in January 2026.

- Designed and built the backend as a single Go deployable organised into 79 modules across five bounded domains — commerce, social, creator, platform and core.
- Chose a modular monolith over microservices and enforced it structurally: modules communicate by command, query and Kafka event, never by reaching into each other's data, so any domain can be extracted as a build change rather than a rewrite.
- Defined the database once as ent schemas with Atlas generating versioned migrations, removing drift between the ORM and the database across a schema this size.
- Built the trust engine — three-tier verification, escrow with dispute windows and automated release timers, and account-level risk scoring — as explicit state machines.
- Implemented money as minor units in a package that carries currency and refuses cross-currency arithmetic, so escrow, payouts, commissions and refunds round in one place.
- Ran a separate Python/FastAPI service for recommendation, ranking, moderation and forecasting, so retraining a model never requires redeploying the payment path.
- Owned deployment: Docker, Kubernetes, Terraform, Traefik and monitoring, versioned alongside the code.

Stack: Go · chi · ent + Atlas · PostgreSQL · Redis · Kafka · Python · FastAPI · Docker · Kubernetes · Terraform · Linux

Backend Engineer

2025 — present

Newdich Technology

Backend engineering in Node.js and TypeScript as part of the Newdich Technology team, a software, IoT and cybersecurity company operating out of Abuja and Okitipupa.

- Build backend services in Node.js and TypeScript on a shared, team-owned codebase — service design, data modelling and API implementation.
- Define typed API contracts between services rather than shared internals, so a boundary stays legible to engineers who did not write it.
- Designed and built the Newdich Technology company website, governed by a written design specification the code must follow.

Stack: Node.js · TypeScript · REST APIs · PostgreSQL · Docker · Linux

— SELECTED PROJECTS

Zudojs

Author · open source

A 39-package modular TypeScript framework — the building blocks under a backend, rather than another HTTP framework.

- Independent packages for DI, lifecycle, configuration, logging, events, CQRS, HTTP, database, transactions, queues, messaging, storage, security, tenancy, feature flags and observability.
- Token-based dependency injection instead of decorator reflection, and lifecycle as a real state machine — startup ordered by declared dependencies, teardown in reverse.
- A CLI that generates fullstack projects, plus 11 frontend framework adapters over the same backend contracts; the foundation under BetNG and Scriptune.

Stack: TypeScript · Node.js · pnpm workspaces · Zod · ESM

Documentation: zudojs.oyinlola.site Source: github.com/oyinlola-tech/zudo

BetNG

Author · open source

A virtual football platform where every match is simulated once and seen identically on five clients, with twelve TypeScript and Python services behind one gateway.

- Built 12 backend services — a gateway and seven TypeScript services on Zudojs and Prisma, plus four Python services on FastAPI — each writing only its own PostgreSQL schema and calling the others over authenticated RPC.
- Made the match result tamper-proof by design: betting closes before kick-off, the seed is an HMAC under a server secret, and database triggers reject any change to a committed result.
- Five clients (React 19 web, TV, shop and admin; Expo mobile) over a 187-route gateway, integer-kobo money end to end, and 14 invariants each mapped to the constraint and test that prove it — 2,200+ Python tests and 171 TypeScript test files.

Stack: TypeScript · Zudojs · Prisma · Python · FastAPI · PostgreSQL · Redis · React · Expo · Playwright · Docker

Source: github.com/oyinlola-tech/BetNg

SentinelX

Author · open source

A self-hosted network intrusion detection and prevention platform where every alert shows its evidence, thresholds and risk breakdown.

- 13 detectors, a bounded YAML rule language with embedded tests, and 7 kill-chain patterns correlating detections into incidents with explained 0–100 risk scores.
- Guarded prevention across 4 firewall adapters (nftables, iptables, pf, Windows Firewall): off by default, enabled only by a typed confirmation, behind a safety guard that refuses to block protected addresses.
- 100% detection with zero false positives across 13 synthetic attack experiments in the committed benchmark; 1,876 tests across unit, API, integration and kernel suites.

Stack: Python · FastAPI · SQLAlchemy · PostgreSQL · Redis · Scapy · Next.js · Docker · nftables

Source: github.com/oyinlola-tech/SentinelX

CommitGuard

Author · open source

Blocks AI-agent attribution in Git commit metadata — in local hooks, a GitHub Action and a webhook-driven GitHub App — from one detection engine.

- No misjudged cases on a 9,174-case labelled dataset, reached after four of 15 recorded benchmark runs exposed look-alike and invisible-character bypasses; every failing run is kept in the repository.
- Server-side checks read policy from the pull request's base commit, so a change cannot relax the rules judging it; every failure path blocks rather than passes.
- 1,452 Python tests, 354 of them security regressions; CI on Linux, macOS and Windows; published to PyPI and the GitHub Marketplace.

Stack: Python · Typer · Pydantic · SQLite · GitHub Apps · GitHub Actions · React · TypeScript · Hypothesis · Playwright

Source: github.com/oyinlola-tech/commitguard PyPI: pypi.org/project/commitguardian Marketplace: github.com/marketplace/actions/commitguard-ai-attribution-check

Kolo

Backend engineer

Digital infrastructure for Ajo and Esusu cooperative savings groups, with Nomba as the payment rail.

- Double-entry ledger: balances are derived from entries, so a discrepancy between what a member paid and what the group holds is reconstructable rather than lost.
- Hardened webhook path — HMAC signature verified first, raw event persisted, then an independent out-of-band re-verification before any wallet is credited.
- 18 controllers, 30+ services and 32 Prisma repositories on Fastify 5, with 14 BullMQ queues keeping verification, payouts and notifications off the request path.

Stack: TypeScript · Fastify 5 · Prisma · PostgreSQL 15 · Redis 7 · BullMQ · React 19 · Nomba · Argon2

Source: github.com/oyinlola-tech/kolo-nomba

Scriptune

Author · open source

Hear a hymn or a Bible verse and find out what it is: one API behind a website, an iPhone app and an Android app, with self-hosted Whisper speech recognition.

- Zudojs modular-monolith API with 9 modules, 51 routes and 29 Prisma models over PostgreSQL 17, documented with OpenAPI.

- Three-stage recognition search — full-text, then pairs of the rarer words, then pg_trgm similarity — so the expensive fuzzy stage runs only when the cheap index lookups find nothing.
- 7 Bible translations including Yoruba and 1,200 hymns, offline on the phone through SQLite FTS5 and on-device whisper.rn transcription.

Stack: TypeScript · Zudojs · Prisma · PostgreSQL · Python · FastAPI · Whisper · Next.js · Expo · React Native

Source: github.com/oyinlola-tech/scriptune

Telente CBT

Lead engineer

Multi-tenant computer-based testing platform for schools and universities, with AI-assisted question generation.

- 47 Prisma models covering the exam, school and billing domains on Fastify over PostgreSQL 16 and Redis 7.
- A Python/FastAPI AI service reachable only from the TypeScript backend behind a service key — no client request ever reaches the model layer directly.
- Timed exams with per-candidate shuffling across four question types, with session state in Redis to keep per-second writes off the transactional path.

Stack: TypeScript · Fastify · Prisma · PostgreSQL 16 · Redis 7 · Python · FastAPI · Docker · Nginx

Church Management System

Author · open source

Records system and public website for a church in Okitipupa, Ondo State.

- Express 5 and MySQL API with 70 endpoints over 16 tables, covering members, households, attendance, finance and public content.
- Four roles mapped to 23 permissions and checked on all 53 admin endpoints; revocable JWT sessions and a write-level audit log.
- 40 automated tests: the real router stack against a database double, and every HTML page's scripts run under linkedom.

Stack: Node.js · Express 5 · MySQL · JWT · Multer · Tailwind CSS

Source: github.com/oyinlola-tech/cms

PowerWatch

Backend engineer · team project

Crowd-sourced electricity outage tracking for Nigeria — the backend of a team project for the Orange internship programme.

- 62 API endpoints over 20 Prisma models, including a six-level geographic hierarchy — country through neighbourhood — so one dataset answers both street-level and state-level questions.
- Each report opens, joins or closes its neighbourhood's outage inside one transaction, with durations recorded in minutes and daily, weekly and monthly rollups materialised on demand.

Stack: TypeScript · Fastify · Prisma · MySQL · Firebase · Swagger

Source: github.com/circorangeintern/Prism-circle

All 30 case studies, with architecture and trade-offs: oyinlola.site/work

— EDUCATION

BSc Computer Science

University of the People · In progress

Strengthening foundations in software engineering, computer systems, programming, databases and CS theory.

Senior Secondary School Certificate

Optimum Victory College, Okitipupa, Ondo State · Completed

— CERTIFICATIONS

Front-End Development — ALX · *Practical web development fundamentals.*

AI Career Essentials — ALX · *AI concepts and AI-assisted workflows.*

Virtual Assistant Certification — ALX · *Remote work, digital productivity and professional workflows.*